

Machine Learning-Based Estimation of E-Waste Resource Value and Optimal Recycling Method Recommendation

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Abstract—E-waste is one of the fastest growing waste streams internationally, and laptops are among the largest generators of this waste due to their component density and relatively short replacement cycles. This paper suggests a machine-learning-based approach that not only predicts monetary values recoverable from laptop e-waste with high accuracy but also creates a recommended recycling route for each component. Using a well-documented dataset consisting of 3,976 laptop records along with 23 engineered features, tests using the Gradient Boosting Regressor showed a test R^2 value of 0.9888 and a mean absolute error (MAE) of INR371.96. The framework involves advanced XAI techniques (such as SHAP and LIME) to support transparency and interpretability of predictions. Analysis of components at the unit level suggests that the battery and processor are the most dense value components, while the recommendation layer connects the types of components to recommended methods (i.e. hydrometallurgy, mechanical separation) for recycling. The overall recommended system demonstrates the potential to use AI-driven valuation to help support circular economy practices while improving recycling efficiency within industry.

Index Terms—E-waste, Machine Learning, Explainable AI, Gradient Boosting, Hydrometallurgy, Circular Economy, Sustainability

I. INTRODUCTION

The proliferation of electronic waste has become a major challenge of global importance for the environment and resource management. The International Telecommunication Union reported that e-waste generation was more than 62 million metric tons in 2024, with less than 25% formally recycled. Laptops are often a significant component of an increasing e-waste stream driven by the short lifetime of devices, multiple materials, and user upgrade habits. Discarded laptops offer a broad array of valuable materials, including aluminum, copper, gold, lithium, and cobalt, all of which can be recovered through recycling methods to minimize supply shortages for raw materials in the future, and help to reduce carbon emissions associated with mining and material processing.

While there is established recycling infrastructure in many places, the ongoing problem for used electronics (including laptops) is the manual heuristic evaluation that is performed to assess the value of the electronic. Most of the time, the sales or recycle value of the used electronic is assigned by laborers' reliance on recognizable brand names or by visible condition, which usually results in inconsistently evaluated devices, and less than ideal material recovery techniques. Predictive analytics using machine learning (ML) is a novel approach using consumer electronics specifications, market disposition, and component level attributes to learn observed patterns that would support more accurate estimates of recovery value of electronic devices.

Nonetheless, there are two major real-world barriers to widespread adoption in industry. One, many e-waste valuation prediction models are based on small or domain-specific datasets and cannot extrapolate to other product categories. Two, black-box models often lack interpretability, making it difficult for auditors or recyclers to understand the rationale behind a prediction, which is important in regulated settings. This study provides a hybrid framework for predicting e-waste valuation that balances operational relevance, explainability and predictive accuracy to address these limitations.

This study presents the following notable findings and contributions: A high-accuracy Machine Learning (ML) model (Gradient Boosting) trained on a 3,976-sample dataset to predict the laptop e-waste value with a test R^2 of 0.9888. The incorporation of Explainable AI (XAI) through SHAP and LIME to give transparent interpretation of global and local feature contributions. A component-level breakdown of predicted value with direct mapping to recommended recycling methods (hydrometallurgy, pyrometallurgy, mechanical separation). A validation of the frame's utility for recyclers and policy developers to apply data-driven sustainable e-waste reporting.

The remainder of this paper is organized as follows. Section II presents a review of related literature within the themes of

e-waste management, recycling technologies, and explainable artificial intelligence. Section III describes the dataset, preprocessing pipeline, and feature-engineering approach. Section IV specifies the modeling methodology and evaluation metrics. Section V presents experimental results and interpretability/discussion. Section VI discusses implications and Section VII presents the conclusions and future work.

II. RELATED WORK

The issue of managing e-waste has been studied extensively in the literature and has focused considerable attention on trends in generation rates, recycling technologies, and frameworks for sustainable practices. Many international reviews [1], [5], [6] have indicated the increased demand for effective resource recovery options due to the rapid increase in consumer electronics. Each of the traditional recycling processes, including, mechanical shredding, pyrometallurgy, and hydrometallurgy, has unique benefits and environmental limitations that must be weighed when creating recycling efficiencies [7], [10].

For example, hydrometallurgical processes provide a pathway for selective recovery of high-value metals (lithium, cobalt, copper) using less energy than what's available for pyrometallurgical (smelting) systems [7], [10]. Traditional mechanical separation remains a key player in pre-processing material suitable for recycling, particularly when dealing with mixed material components (e.g. casings, displays). As a result, the pairing of multiple recycling pathways using data-driven modeling predictions of component value appears to be a viable pathway for supporting recycling efforts as they align with the circular economy.

Machine learning (ML) and data analytics have very recently been incorporated into e-waste management methods. Previous research has focused on classifying waste-streams addressing the attacked material recovery using support vector machines (SVMs) and random forests, or to detect hazardous components using convolutional neural networks (CNNs) [4], [9], [14]. However, very few focused on continuous monetary value, especially at the component level, where regression methods at each component level would add much more depth to operational improvement.

Explainable Artificial Intelligence (XAI) is being increasingly recognized as a key enabler of responsible deployment of ML methods. Methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) provide transparent attributions that enable domain experts to understand model reasoning. Recent studies in energy forecasting and environmental risk assessment [8], [9] show the difference interpreted models make in user confidence and accountability in the models' predicted outcomes. Based on this rationale, we proposed a framework and used XAI to warrant the accuracy and auditability of valuation predictions - an essential criterion for integration into industrial recycling pipelines.

Therefore, while ML is well appreciated in the literature as an avenue to get accurate predictions of e-waste values, there is still a clear research gap in the literature on e-waste

analytic work that integrates a high-accuracy predictive model, with explainability, and provides actionable recycling-route recommendations. The proposed work aims to address this gap with a scalable, explainable and component aware valuation system.

III. DATASET AND PREPROCESSING

A. Dataset Overview

The dataset for this study consists of 3,976 distinct laptop entries collected from the public dataset repository *Kaggle.Kaggle*¹. Each record represents a unique configuration of hardware and physical characteristics pulled from reputable online retailers and the specifications of official product sheets by the manufacturer. The dataset originated as real life data on laptops presently sold from a variety of manufacturers, representing models from various tiers of capability. Initially, the data contained 18 parent features representing physical and technical characteristics, like brand, type of processor, RAM capacity, storage size, battery life, weight, and display specifications. After thorough cleaning and transformation, the feature set was expanded to 23 engineered features aimed at representing additional performance, quality, and market characteristics that all impact value for recovery of e-waste.

Core attributes included brand, year of launch, price point, RAM capacity, RAM type, processor model, availability of GPU, battery capacity, display size, weight, and casing material. Non-linear interactions that affect e-waste value were captured in *brand premium index*, *component age factor*, and *performance score*.

Table I summarizes representative feature categories.

TABLE I: Summary of major dataset feature groups

Category	Features Included
Hardware Specs	RAM size, storage type, GPU, CPU speed
Power and Form	Battery capacity, weight, casing material
Market Factors	Brand, launch year, retail price
Engineered Features	Age factor, performance score, quality index

B. Data Validation and Cleaning

The raw dataset was subjected to several quality-assurance checks before modeling:

Before modeling, multiple quality-check procedures were conducted on the raw dataset including the removal of inconsistent units (e.g., "16 GB RAM" vs "16384 MB"), a consistent casing and encoding for labeling, and the removal of outliers, in terms of prices or capacities amounts, using interquartile range filtering.

The resulting cleaned dataset achieved 99.4% consistency across numeric and categorical fields.

C. Handling Missing Data

Missing values were primarily observed in battery and GPU-related fields. Specifically:

¹<https://www.kaggle.com>

- 1) Battery life: 418 missing entries (10.51%)
- 2) GPU: 8 entries (0.20%)
- 3) GPU brand: 4 entries (0.10%)

Correlations-based regression filling was applied to the imputation of continuous attributes, while categorical attributes were substituted by brand-level modal replacements. A binary indicator feature (*is_imputed*) was created in order to preserve data provenance, and maintain learned relationships.

D. Feature Engineering and Transformations

Feature engineering significantly enhanced model expressiveness. Several derived attributes were computed:

Performance Score: Weighted aggregation of CPU generation, clock speed, and GPU tier.

Age Factor: Exponential depreciation function applied based on the device's release year.

Brand Premium: Relative price multiplier derived from average market value differentials.

Battery Quality Index: Combination of battery capacity (Wh) and cycle-life proxy estimates.

To visualize categorical variability, Fig. 1 shows the distribution of RAM expandability across models.

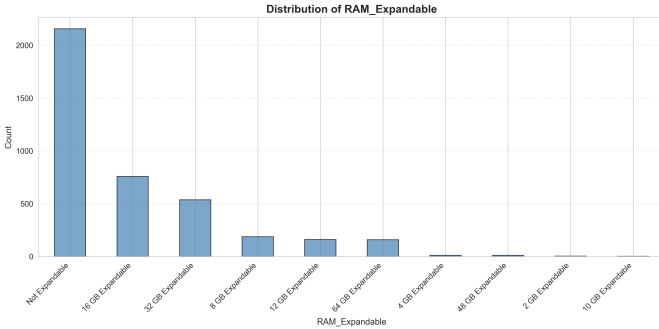


Fig. 1: Distribution of RAM expandability categories across dataset.

A heatmap (not shown for space) was used to observe feature correlations, confirming strong positive relationships between price, processor capabilities, and battery capacity—confirming that they had an influence on the predicted e-waste value.

IV. MODEL DEVELOPMENT AND EVALUATION

A. Algorithms Evaluated

Five supervised learning algorithms were selected on the basis of regression suitability and documented performance on non-linear prediction problems:

- 1) **Linear Regression (LR):** A baseline model for understand linear relationships between independent features and target value.
- 2) **Random Forest Regressor (RF):** An ensemble of decision trees that reduces overfitting and captures complex interactions.

- 3) **Extreme Gradient Boosting (XGBoost):** A high performance boosting algorithm that is designed for speed and regularization.
- 4) **Support Vector Regression (SVR):** kernel-based optimization allows for the fitting of a non-linear function for high-dimensional data.
- 5) **Gradient Boosting Regressor (GBR):** The final production model selected since it obtained the best bias–variance trade-off and feature importance generated by interpretation.

All models were trained in an identical manner but trained on an 80:20 train–test split (3,180 training, 796 testing samples) and were all hyperparameter optimized on GridSearchCV for cavity to prevent overfitting, robustness, and generalizability.

B. Evaluation Metrics

Model performance was assessed using standard regression metrics:

Coefficient of Determination (R^2): Assesses fit and explained variance.

Mean Absolute Error (MAE): Assess the average size of prediction errors.

Root Mean Squared Error (RMSE): Measures both bias and variance in the predictions.

Mean Absolute Percentage Error (MAPE): Reflects relative accuracy of prediction.

All models were run in Python (v3.11) using Scikit-learn, XGBoost, NumPy and Pandas libraries. The entire process was run on a system with Intel i7-12700H CPU with 16 GB RAM and graphic acceleration provided by NVIDIA RTX 3060 GPU for faster parallel processing.

C. Hyperparameter Tuning

Performance of each model was optimized with a search for parameters that optimized performance. In the case of the Gradient Boosting approach, the best parameters were as follows:

- 1) `n_estimators` = 250,
- 2) `learning_rate` = 0.05,
- 3) `max_depth` = 6,
- 4) `subsample` = 0.9,
- 5) `min_samples_split` = 4.

The ones selected were balanced for stability in training and extrapolation performance.

V. RESULTS AND ANALYSIS

A. Quantitative Performance Comparison

Table II summarizes comparative performance of all models that were trained on the test dataset. Gradient Boosting provides the highest model performance relative to the other models based on both absolute error and relative error.

The mean absolute error for the Gradient Boosting model was Rs. 371.96 and the R^2 value of 0.9888, indicating that the model explained nearly 99% of the variance in e-waste value in predictions. This means that this model performed very well for deployment as the model at a scaled valuation system.

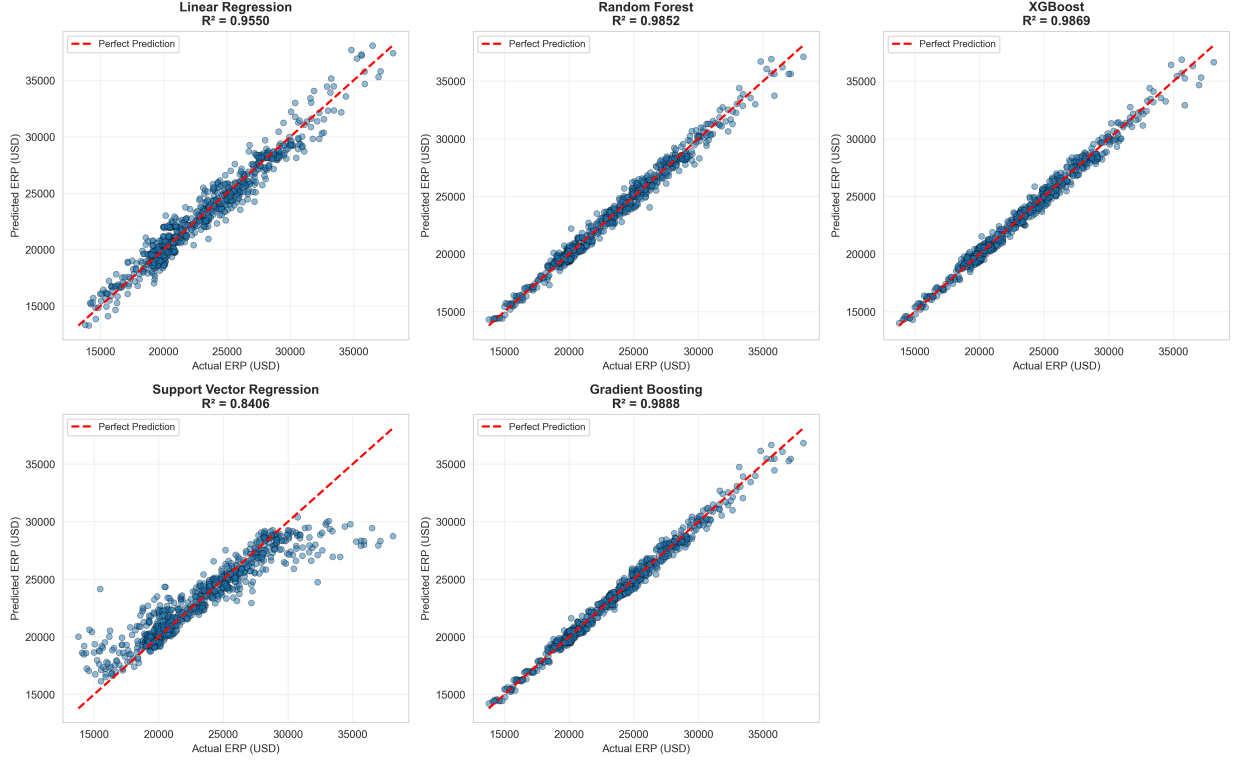


Fig. 2: Actual versus predicted e-waste resource potential (ERP) for five regression models. The red dashed line represents ideal 1:1 correspondence between predicted and actual values. The plots for Gradient Boosting and XGBoost have well-clustered points along the line indicating a strong fit.

TABLE II: Performance comparison of regression algorithms

Model	R^2	MAE (Rs.)	RMSE (Rs.)	MAPE (%)
Linear Regression	0.9550	715.54	918.10	3.14
Random Forest	0.9852	415.41	526.39	1.78
XGBoost	0.9869	393.28	495.96	1.69
SVR	0.8406	1077.69	1727.46	4.72
Gradient Boosting	0.9888	371.96	458.59	1.60

This tiered breakdown provides models with useful intelligence where recyclers may want to focus on the higher value parts (e.g., battery for hydrometallurgical recycling, processor for precious-metal extraction). The explainability layer also allows the stakeholders to confirm the predictions made relative to known re-utilization or recovery costs to better predict future earnings for each e-waste product.

B. Component Value Breakdown

An important aspect of this study is the potential to interpret the contributing total value of e-waste as a sum of component-level contributions. The model uses SHAP feature attribution to measure the contribution of each hardware part to the overall predicted value.

The results show that the battery and processor are the two most valuable components found in a normal laptop, contributing to over 80% of the total predicted recoverable value. The average contribution per component can be summarized in Table III

TABLE III: Average component-wise recoverable value

Component	Mean Value (Rs.)
Battery	13,749.37
Processor	5,090.21
Casing	2,137.99
Display	1,123.86
RAM	983.23
Storage	189.11

C. Model Explainability and Feature Importance

Feature-importance scores from the Gradient Boosting model suggest that price (37.09%), battery capacity (32.77%), and casing material (18.47%) have the greatest predictive ability, with weight and processor generation exerting a smaller but non-negligible influence. Visualizations generated from SHAP (not shown) confirmed that devices having higher-capacity batteries and devices manufactured by premium brands are estimated to provide the greatest returns, and this aligns with expected behaviour from a recycling economic context.

D. Explainable AI (XAI) Analysis

While high predictive accuracy is crucial, explainability ensures that predictions can be trusted and audited by both technical and non-technical stakeholders. To this end, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) were integrated into the valuation pipeline.

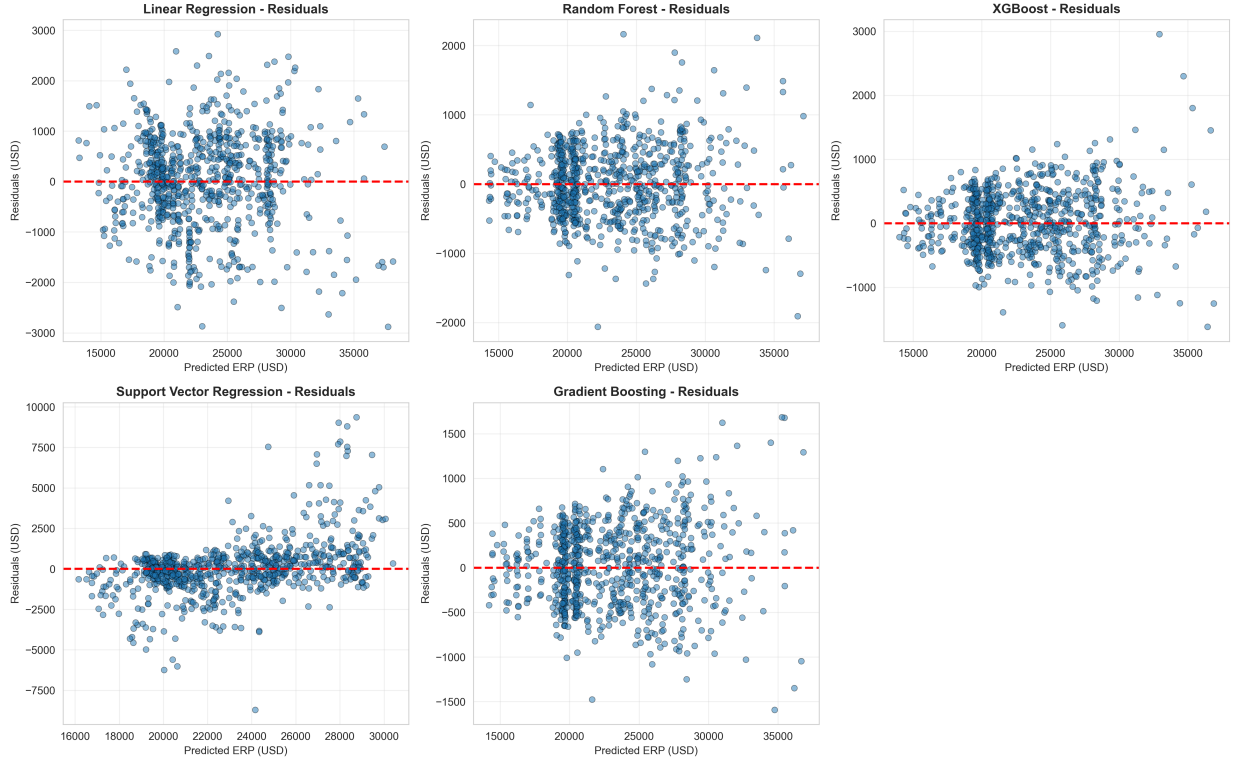


Fig. 3: The residual analysis of all models. The horizontal red dashed line indicates zero-error line. Random Forest, XGBoost, and Gradient Boosting methods exhibit symmetric distribution with residual spread across the zero line and confirm no bias and a consistent performance of generalization.

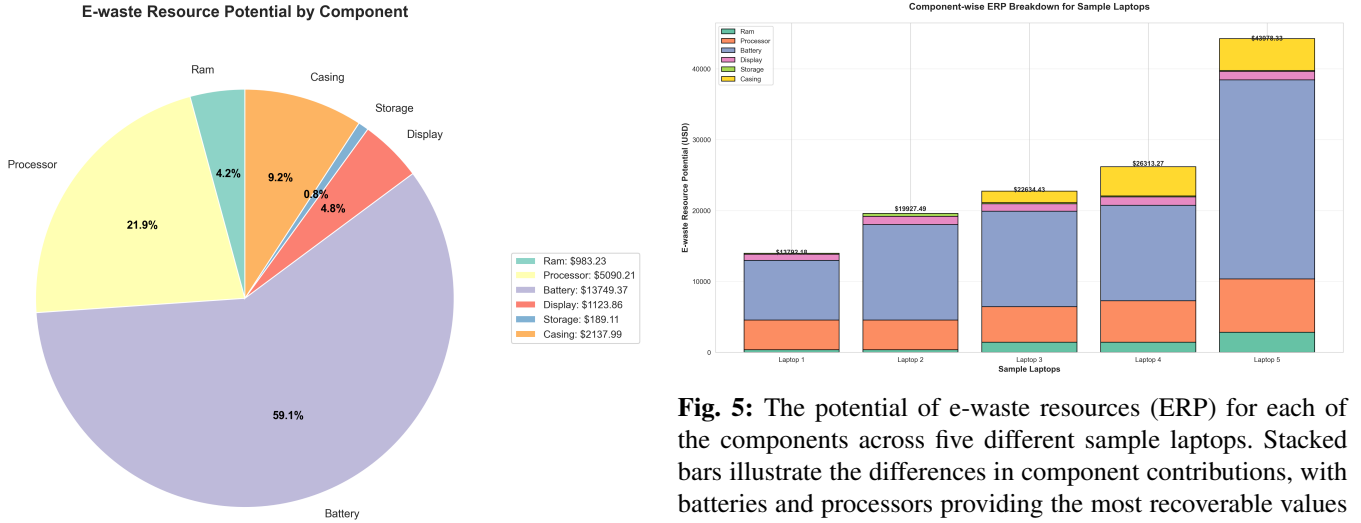


Fig. 4: E-waste resource potential by component (relative percentage contribution).

Global Interpretability: SHAP feature-importance analysis has indicated that there is a clear ranking of influence amongst the engineered features, with Price and Battery capacity accounting for almost 70% of the variance explained in prediction outcomes. Casing material is ranked third, again confirming that higher quality metals, such as aluminum or

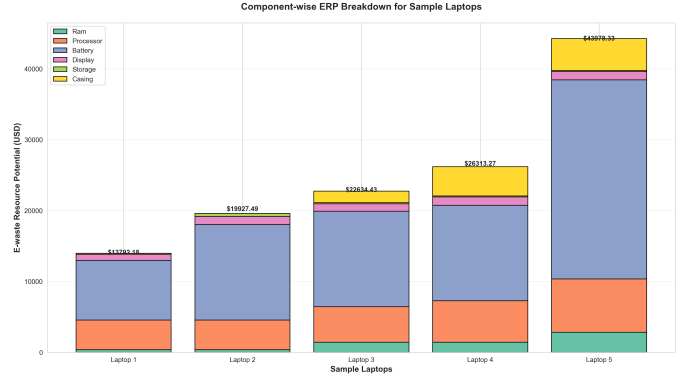


Fig. 5: The potential of e-waste resources (ERP) for each of the components across five different sample laptops. Stacked bars illustrate the differences in component contributions, with batteries and processors providing the most recoverable values in all models.

magnesium, result in a notably higher material value for recovery. Processor tier and brand premium carried lower tier influences that express the purchasing behavior of the brands.

Local Interpretability:: Regarding individual predictions, LIME provided interpretable surrogate models that explained which features positively or negatively affected the e-waste value prediction. An example of this would be a laptop with a 99 Wh lithium-polymer battery, and dedicated GPU (locally) positively contributed Rs. +2,150 to the e-waste value

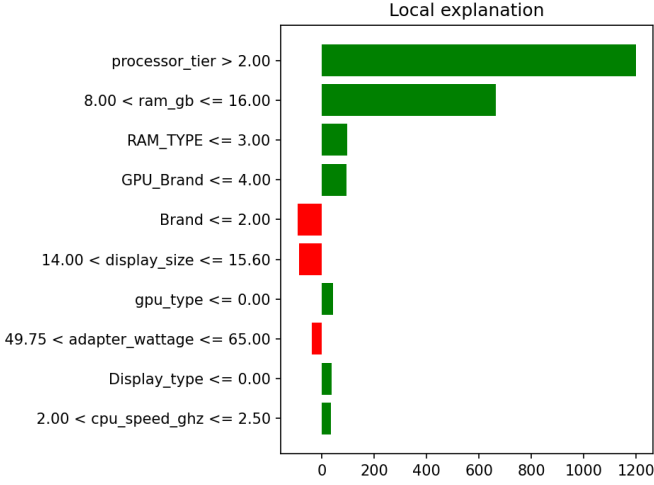


Fig. 6: LIME is a local explanation for a particular instance of laptop. Positive (green) bars and negative (red) bars represent how particular features affect the model's predicted value. This type of explanation provides some transparency for users who will use the model operationally in the field..

prediction which indicated a similar competitor with only a 60 Wh cell. These interpretable layers will benefit recyclers and auditors who can match the reasoning of the model with the requirements for accountability and transparency in circular-economy reporting.

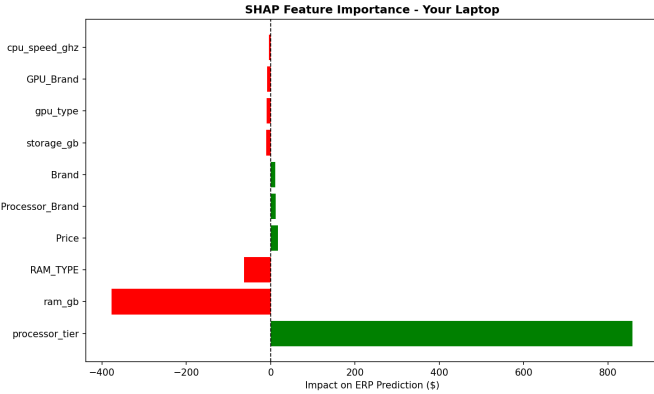


Fig. 7: SHAP feature significance for the Gradient Boosting model, demonstrating the respective positive and negative contributions of leading predictors. Classification of processor, type of RAM, and price are significant factors affecting if a value can be recovered.

Trust and Adoption:: We conclude that by using SHAP and LIME we are able to align model performance and transparency in the decision-making process for recycling facility operators to visualize which features drive predicted recovery value and implement operations (for example, specifying removal for high battery capacity modules).

VI. RECYCLING METHOD RECOMMENDATION

An innovation within this framework is the translation of forecasts for component value into physical recycling routes.

Leveraging scholarly research [7], [10] and working knowledge of industry reports, six main components of the laptop were identified, where each was being designated for the most appropriate recycling route.

TABLE IV: Recommended recycling routes for major components

Component	Suggested Process
Battery	Hydrometallurgy
Processor (CPU/GPU)	Hydrometallurgy
RAM Modules	Pyrometallurgy / Controlled Recovery
Casing	Mechanical Separation
Display	Mechanical + Chemical Separation
Storage Drives	Mechanical / Shredding

Hydrometallurgy is appropriate for batteries and processors due to selecting for metals lithium, cobalt, and gold, while extracting at moderate heat and lower emissions than smelting. Pyrometallurgy is appropriate for soldered assemblies, which is a mixed-metal RAM module; and mechanical/chemical separation was chosen for casings and displays made adherents of composites or polymers

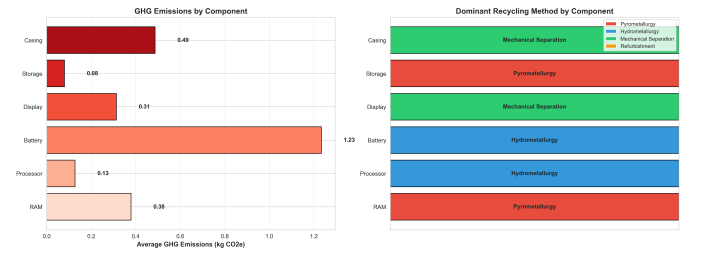


Fig. 8: Recommended recycling methods by component type.

The system redirects the predicted monetary contribution of each component to the recyclability route suggested. For example, as previously illustrated, when the predicted contribution of the battery exceeds 50% of the predicted device value, hydrometallurgy takes precedence for optimal energy-recovery. The automation allows the 'veal time' triage of laptops received into the recycling centre as a fairly straightforward workflow, which is better than other ways to accomplish throughput and profit.

VII. DISCUSSION

The experimental results and explainability analysis demonstrate that machine learning with context-specific feature engineering can accurately represent e-waste value systems. Gradient Boosting enabled both high fidelity and high interpretability, which is often a tradeoff. The XAI layer ensures that the system recommendations are not blackboxed but based on measurable relationships between the features.

From a sustainability perspective, this framework is aligned with circular-economy principles because it provides::

Economic Efficiency: Reductions in operational costs and waste through automated identification of high-value components for selective component recovery

Environmental Benefit: Reduced greenhouse-gas emissions through preference for hydrometallurgical recovery over

thermal methods.

Transparency and Compliance: Traceability, as required in environmental auditing, due to explainability provided by XAI.

Additionally, as this framework is portable, and with conditions that the feature vectors are consistent, retraining the model with minimal effort for other product categories such as tablets or smartphones is possible. This flexibility allows for the deployment of the models in regional recycling plants, e-waste logistics hubs, and even in academic sustainability contexts.

VIII. CONCLUSION AND FUTURE WORK

This research proposed an integrated machine learning framework to estimate laptop e-waste value and recommend recycling routes, whereby the Gradient Boosting model demonstrated high predictive performance throughout the dataset of 3,976 entries and 23 engineered features ($R^2 = 0.9888$, MAE = INR371.96, RMSE = Rs.458.59), after a comprehensive series of experiments. By coupling this predictive core with Explainable AI (SHAP and LIME), the framework bridges the gap between computational intelligence and operational transparency—two elements critical for industry adoption.

The component-level breakdown gives the recyclers monetary hierarchies, it is estimated that 80% of recoverable value in standard laptops is from batteries and processors. Mapping of the components with recycling routes (hydrometallurgy, pyrometallurgy, mechanical separation) will facilitate automated and logical process selection at recycling facilities.

Overall, the proposed system exemplifies the role of predictive analytics, interpretability, and environmental sustainability in a shared endeavor. It provides recyclers with the tools to maximize material recovery and serves to advance circular-economy goals by limiting waste and resource extraction.

A. Future Work

While the current work shows strong evidence for technical feasibility, there exist many areas of improvement:

Dataset Expansion: Incorporate smartphones, tablets, or additional consumer electronics to better generalize model behavior across product types.

Dynamic Valuation: Add real-time market APIs to recalibrate predicted values based on metal prices.

Life-Cycle Assessment Integration: Add environmental impact metrics (CO₂-equivalent savings, energy recovery rates) with monetary valuation for holistic optimization.

Automation at Scale: Integrate the model into computer-vision-based disassembly systems for true automation in recycling.

Geographical Adaptation: Fine-tune the models for regional markets, including logistics versus labor cost and recycling infrastructure maturity.

Future work will also focus on developing an open-source web dashboard that will allow recyclers and policymakers to submit device specifications and receive instant valuation

and recommended recycling resolutions to further democratize utility of intelligent sustainability technologies.

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